

8

Nutrition and Hydration



1. Identify the six basic nutrients and explain MyPlate

Define the following terms:

nutrient

a necessary substance that provides energy, promotes growth and health, and helps regulate metabolism.

nutrition

how the body uses food to maintain health.

1. Identify the six basic nutrients and explain MyPlate

REMEMBER:

A well-balanced diet will help residents maintain muscles and skin tissues and prevent pressure ulcers.

Transparency 8-1: Six Basic Nutrients

- Water
- Carbohydrates
- Protein
- Fats
- Vitamins
- Minerals

1. Identify the six basic nutrients and explain MyPlate

Important facts about water:

- Most essential nutrient for life
- Water aids in
 - Digestion
 - Absorption of food
 - Elimination of wastes
 - Maintaining normal body temperature

1. Identify the six basic nutrients and explain MyPlate

Important facts about carbohydrates:

- Provide energy and extra protein
- Provide fiber
- Complex carbohydrates include bread, cereal, potatoes, rice, pasta, vegetables, and fruits
- Simple carbohydrates include sugars, sweets, syrups, and jellies

1. Identify the six basic nutrients and explain MyPlate

Important facts about protein:

- Essential for tissue growth and repair
- Provides a supply of energy
- Protein includes seafood, poultry, meat, eggs, milk, cheese, nuts, nut butters, peas, dried beans or legumes, and vegetarian meat substitutes.

1. Identify the six basic nutrients and explain MyPlate

Important facts about fats:

- Help the body store energy
- Add flavor to food
- Important for absorption of certain vitamins
- Monounsaturated and polyunsaturated are healthier fat. Saturated and trans fats can increase cholesterol and the risk of some diseases.
- Some fats come from animal sources, such as butter, beef and dairy products. Some fats come from plant sources such as olives and nuts.

1. Identify the six basic nutrients and explain MyPlate

Important facts about vitamins:

- Vitamins are essential to body functions.
- Fat-soluble vitamins are A, D, E, and K.
- Water-soluble vitamins are B and C.

1. Identify the six basic nutrients and explain MyPlate

Important facts about minerals:

- Minerals form and maintain body functions.
- Include zinc, iron, sodium, potassium, calcium, magnesium, and phosphorus

Handout 8-1: Vitamins and Minerals

VITAMIN	SOURCE	FUNCTION
Vitamin A	dark green and yellow vegetables, such as broccoli and turnips	assists with skin and eye development; keeps the skin healthy; helps the eyes adjust to dim light; helps the linings of the respiratory and digestive tracts resist infection
Vitamin C	fruits such as oranges, strawberries, grapefruit, and cantaloupe; vegetables such as broccoli, cabbage, brussels sprouts, and green peppers	assists with healing wounds and building bones and teeth; holds cells together; strengthens the walls of blood vessels; and helps the body absorb iron
Vitamin B2 or riboflavin	milk, milk products, lean meat, green leafy vegetables, eggs, breads, and cereals	helps cells use oxygen, which allows them to release energy from food; important for protein and carbohydrate metabolism; needed for growth, healthy eyes, skin, and mucous membrane

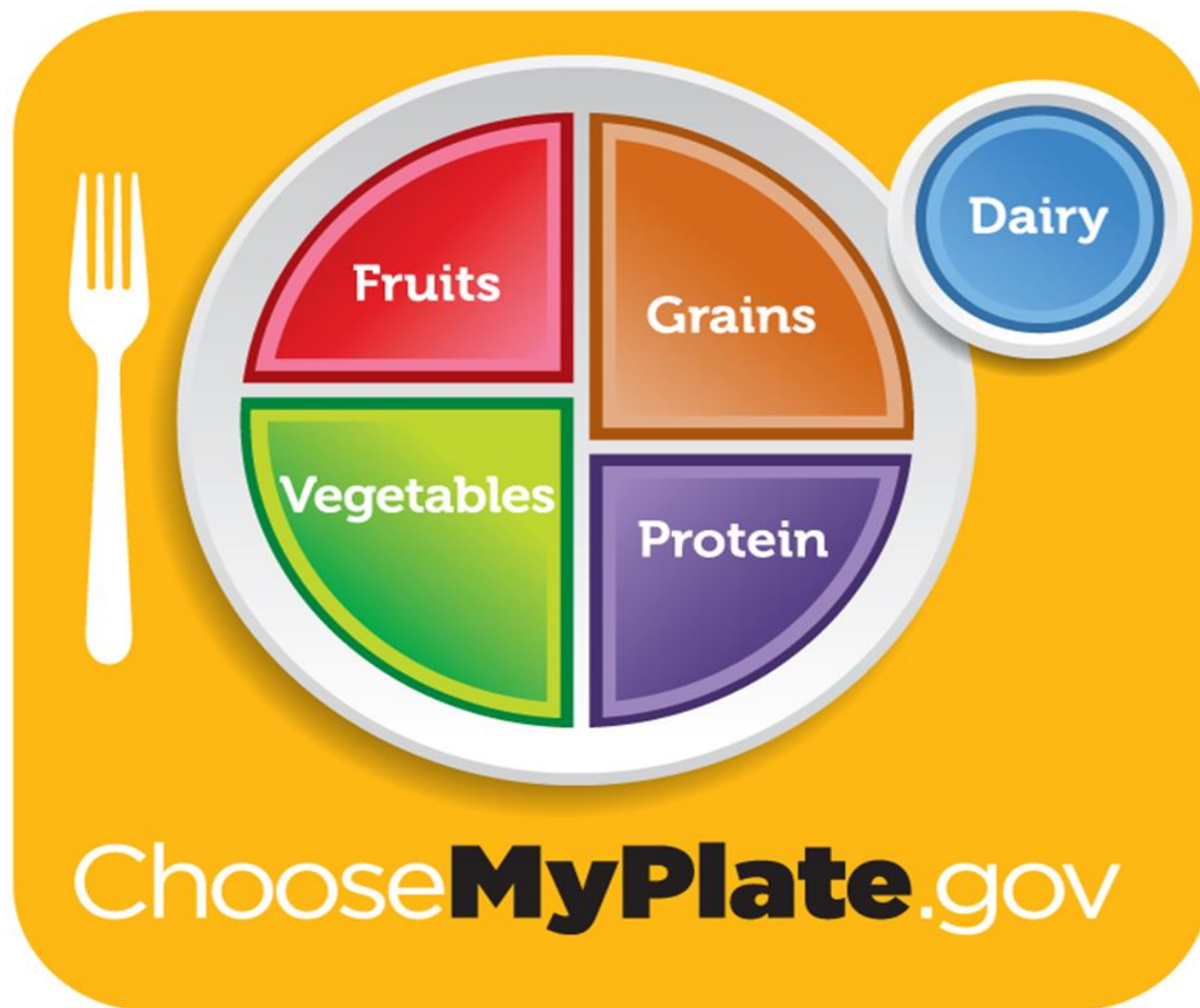
Handout 8-1: Vitamins and Minerals (cont'd)

VITAMIN	SOURCE	FUNCTION
Vitamin B3 or niacin	lean meat, poultry, fish, peanuts and peanut butter, whole grain breads and cereals, peas, beans, and eggs	important for protein, carbohydrate, and fat metabolism; important for appetite; important for the functioning of the skin, tongue, nervous system, and digestive system; helps cells use oxygen for energy
Vitamin D	milk, butter, liver, and fish liver oils; also obtained by exposing the body to direct sunlight, which interacts with the cholesterol in the skin	responsible for the body's absorption of the minerals calcium and phosphorus and contributes to the formation of healthy bones; especially important to growing children and women who are pregnant or breastfeeding
Vitamin E	cereals, nuts, vegetable oils, wheat germ, vegetables, fish, and fruits	helps cells use oxygen, which allows them to release energy from food; important for protein and carbohydrate metabolism; needed for growth, healthy eyes, skin, and mucous membrane
Thiamin	lean pork, dried beans, peas, whole grain and enriched breads and cereals, and certain types of nuts	helps the body obtain energy from foods

Handout 8-1: Vitamins and Minerals (cont'd)

VITAMIN	SOURCE	FUNCTION
Iron	egg yolks, green leafy vegetables, breads, cereals, and organ meats	necessary for red blood cells to carry oxygen; helps in the formation of enzymes
Sodium	almost all foods and table salt	important for maintaining fluid balance (helps the body retain water)
Calcium	milk and milk products, such as cheese, ice cream, and yogurt; green leafy vegetables, such as collards, kale, mustard, dandelion, and turnip greens; and canned fish with soft bones, such as salmon	important for the formation of teeth and bones, the clotting of blood, muscle contraction, and heart and nerve function
Potassium	fruits and vegetables, cereals, coffee, and meats	essential for nerve and heart function and muscle contraction
Phosphorus	milk, milk products, meat, fish, poultry, nuts, and eggs	needed for the formation of bones and teeth and for nerve and heart function; important for the body's utilization of proteins, fats, and carbohydrates

Transparency 8-2: USDA MyPlate



1. Identify the six basic nutrients and explain MyPlate

NAs should remember these points about vegetables and fruits:

- Half a person's plate should consist of fruits and vegetables.
- Dark green, red, and orange vegetables have the best nutritional content.
- Vegetables are low in fat and calories and have no cholesterol.
- Vegetables provide fiber and vitamins.
- Fruits are low in fat, sodium, and calories and have no cholesterol.
- Fruits provide vitamins and fiber.

1. Identify the six basic nutrients and explain MyPlate

NAs should remember these points about grains:

- At least half of all grains consumed should be whole grains.
- Whole grains contain the bran and germ, as well as the endosperm. Refined grains retain only the endosperm.
- Grains are found in cereal, bread, rice, and pasta.

1. Identify the six basic nutrients and explain MyPlate

NAs should remember these points about proteins:

- Meat, poultry, seafood, and eggs are animal sources of proteins. Beans, peas, soy products, nuts, and seeds are plant sources of proteins.
- Eat seafood twice a week instead of meat.
- Choose lean meat and poultry. Include eggs and egg whites on a regular basis.
- Eat plant-based protein foods often.
- Some nuts and seeds (flax, walnuts) are excellent sources of essential fatty acids.

1. Identify the six basic nutrients and explain MyPlate

NAs should remember these points about dairy:

- Provides protein, vitamins, and minerals
- Includes all of the foods made from milk that retain their calcium content, such as yogurt and cheese
- Most dairy group choices should be fat-free or low-fat (1%).
- Choose fat-free or low-fat milk or yogurt more often than cheese.
- Soy products enriched with calcium are an alternative to dairy foods.

1. Identify the six basic nutrients and explain MyPlate

NAs should remember these tips for making healthy food choices:

- Balance calories.
- Enjoy your food but eat less.
- Avoid oversized portions.
- Eat these foods more often: vegetables, fruits, whole grains, and fat-free or 1% milk and low-fat dairy products.
- Eat these foods less often: foods high in solid fats, added sugars, and salt. These foods include fatty meats, like bacon and hot dogs, cheese, fried foods, ice cream, and cookies.
- Compare sodium content in foods. Select canned foods that are labeled sodium-free, very low sodium, low-sodium, or reduced sodium.
- Drink water instead of sugary drinks.

2. Describe factors that influence food preferences

REMEMBER:

Residents have a legal right to make choices about their food and to refuse food. NAs must honor residents' beliefs and preferences.

2. Describe factors that influence food preferences

Think about this question:

Do you have any regional, cultural, or religion-based food preferences?

3. Explain special diets

Define the following terms:

special, modified, or therapeutic diets

diets for people who have certain illnesses, conditions or food allergies.

puree

to blend or grind food into a thick paste of baby food consistency.

3. Explain special diets

The following are all types of special diets NAs may encounter in LTC facilities:

- Low-sodium diet
- Fluid-restricted diet
- Low-protein diet
- Low-fat diet
- Modified calorie diet
- Diabetic diet
- Vegetarian diet
- Liquid diet
- Soft and mechanical soft diet
- Pureed diet

3. Explain special diets

Think about this question:

Are you already familiar with any of these special diets?

3. Explain special diets

Residents may be advised to add a high-nutrition supplement to a regular or modified diet. NAs should know these facts about supplements:

- Usually added to diet to encourage weight gain or intake of proteins, vitamins, or minerals
- May come in powdered or liquid form
- NAs should follow facility policy about mixing supplements; they may not be allowed to do so
- NAs can help by patiently encouraging residents to drink their supplements
- A resident has the right to refuse a supplement; refusal should be reported to the nurse

4. Describe how to assist residents in maintaining fluid balance

Define the following terms:

NPO

Abbreviation meaning nothing by mouth; medical order that means a person should not have anything to eat or drink.

dehydration

a serious condition that results from inadequate fluid in the body.

4. Describe how to assist residents in maintaining fluid balance

Fluid intake is important because it:

- Helps prevent constipation and incontinence
- Dilutes wastes and flushes out urinary system
- May help prevent confusion

4. Describe how to assist residents in maintaining fluid balance

REMEMBER:

A resident with an order for NPO (nothing by mouth) cannot have any food or drink. An NA should never offer food or drink - not even water - to a resident who is designated NPO.

4. Describe how to assist residents in maintaining fluid balance

The following guidelines can help prevent dehydration:

- Report warning signs immediately.
- Encourage residents to drink every time you see them.
- Offer fresh water and fluids often.
- Offer other forms of liquids if permitted (e.g., ice chips, frozen flavored ice sticks, gelatin).
- Offer sips of liquids between bites of food.
- Make sure the pitcher and cup are close by and are light enough for resident to lift.
- Offer assistance.
- Record fluid intake & output.

4. Describe how to assist residents in maintaining fluid balance

Signs and symptoms of dehydration to observe and report include

- Drinking fewer than six 8-ounce glasses of liquid per day
- Drinking little or no fluids at meals
- Needing help drinking from a cup
- Having trouble swallowing liquids
- Having frequent vomiting, diarrhea, or fever

4. Describe how to assist residents in maintaining fluid balance

Signs and symptoms of dehydration (cont'd):

- Being easily confused or tired
- Resident has any of the following:
 - Dry mouth
 - Cracked lips
 - Sunken eyes
 - Dark urine
 - Strong-smelling urine
 - Weight loss
 - Complaints of abdominal pain

4. Describe how to assist residents in maintaining fluid balance

REMEMBER:

If an NA approaches residents positively, with a question like “Would you like water or juice?” rather than “Do you want anything to drink?” the resident is more likely to stay well hydrated.

NAs should make an effort to find out what residents’ favorite beverages are and offer these at least three times a day, in addition to the beverages served at meals.

Serving fresh water

Equipment: water pitcher, ice scoop, cup, straw, gloves

1. Identify yourself by name. Identify the resident by name.
2. Wash your hands.
3. Put on gloves.
4. Scoop ice into the water pitcher without touching the ice scoop to the pitcher. Add fresh water. Do not touch the pitcher to the spout or faucet.
5. Use and store ice scoop properly. Do not allow ice to touch your gloved hand and fall back into container. Place scoop in the proper receptacle after each use.
6. Take pitcher to resident.

Serving fresh water

7. Pour water into the resident's cup. Offer the resident a drink of water. Leave pitcher and glass at the bedside.
8. Make sure that pitcher and cup are light enough for resident to lift. Leave a straw if the resident desires it and does not have swallowing problems.
9. Place call light within resident's reach.
10. Remove and discard gloves.
11. Wash your hands.

4. Describe how to assist residents in maintaining fluid balance

Define the following terms:

fluid overload

a condition that occurs when the body cannot handle the amount of fluid consumed.

edema

swelling caused by excess fluid in body tissues.

4. Describe how to assist residents in maintaining fluid balance

Signs and symptoms of fluid overload include

- Swelling of extremities
- Weight gain
- Decreased urine
- Shortness of breath
- Increased heart rate
- Anxiety
- Tight, smooth, or shiny skin

5. List ways to identify and prevent unintended weight loss

REMEMBER:

Unintended weight loss is a serious problem for the elderly. It can lead to skin breakdown and/or malnutrition.

5. List ways to identify and prevent unintended weight loss

NAs should remember these guidelines for preventing unintended weight loss:

- Report observations and warning signs to the nurse.
- Food should look, taste, and smell good.
- Encourage residents to eat; talk positively about food.
- Honor food likes/dislikes.
- Offer different kinds of foods and beverages.
- Help residents who have trouble self-feeding.

5. List ways to identify and prevent unintended weight loss

Guidelines for preventing unintended weight loss (cont'd):

- Season foods to residents' preferences.
- Allow plenty of time to finish eating.
- Tell nurse if residents have trouble with utensils.
- Record meal/snack intake.
- Give oral care before and after meals.
- Position residents upright for feeding.
- If resident has a loss of appetite or seems sad, ask about it.

5. List ways to identify and prevent unintended weight loss

NAs should observe for and report the following regarding a resident's appetite:

- Resident needs help eating or drinking
- Resident eats less than 75% of food
- Resident has mouth pain
- Resident's dentures do not fit
- Resident has difficulty chewing or swallowing
- Resident coughs or chokes while eating
- Resident is sad, has crying spells, or withdraws
- Resident is confused, wanders, or paces

6. Identify ways to promote appetites at mealtime

REMEMBER:

Meals are not only a time for getting nourishment but a time for socialization as well. Making mealtime pleasant for residents is one of the most important things an NA can do to help residents get proper nutrition.

6. Identify ways to promote appetites at mealtime

NAs should remember these guidelines for promoting appetites:

- Assist with grooming/hygiene tasks before dining as needed.
- Give oral care before eating if requested.
- Offer a trip to the bathroom or help with elimination needs before eating.
- Help residents wash their hands before eating.
- Encourage use of dentures, eyeglasses, and hearing aids.
- Check the environment. Address odors. Keep noise level low. Do not shout or bang plates or cups.

6. Identify ways to promote appetites at mealtime

Guidelines for promoting appetites (cont'd):

- Seat residents next to friends.
- Properly position residents for eating, which is normally in the upright position.
- Serve food promptly to maintain the correct temperature.
- Plates should look appetizing.
- Give the resident proper eating tools, including assistive utensils if needed.
- Be cheerful, positive, and helpful.
- Give additional food when requested.

7. Demonstrate how to assist with eating

When assisting residents with eating NAs should remember the following:

- Residents will need different levels of help.
- Some residents will only need help with setting up but can feed themselves.
- Some residents will need to be fed. Be sensitive and give privacy.
- Encourage residents to do what they can.

7. Demonstrate how to assist with eating

Guidelines for assisting a resident with eating are as follows:

- Wash your hands.
- Identify residents before serving meals.
- Sit at resident's eye level.
- Allow time for prayer if the resident wishes.
- Be supportive and encouraging. Do not treat the resident like a child.
- Do not touch food to test its temperature. Use a hand over the dish instead.
- Cut foods and pour liquids as needed.

7. Demonstrate how to assist with eating

Guidelines for assisting a resident with eating (cont'd):

- Identify foods and fluids that are in front of resident. Call pureed food by the correct name.
- Ask the resident what he wants to eat first. Allow resident to make the choice.
- Do not mix foods unless resident prefers it.
- Do not rush the meal.
- Be social and friendly. Converse if the resident wishes to do so.
- Give the resident full attention.
- Alternate food and drink, cold and hot, and bland and sweets.
- Honor requests for different food.

7. Demonstrate how to assist with eating

REMEMBER:

An NA should not insist that a resident use a clothing protector and should never refer to a clothing protector as a bib.

<https://videos.hartmanonline.com/SkillsVideos/Feeding/index.php?school=amtrain322>

Feeding a resident

Equipment: meal; eating utensils; clothing protector; washcloths, wipes, or towel

1. Identify yourself by name. Identify the resident by name.
2. Wash your hands.
3. Explain procedure to resident. Speak clearly, slowly, and directly. Maintain face-to-face contact whenever possible.
4. Provide for the resident's privacy with a curtain, screen or door.
5. Look at the diet card or menu. Ask resident to state her name. If the resident is unable to state her name, check identification another way such as looking at a photo ID or an armband. Verify that resident has received the right tray.

Feeding a resident

6. Raise the head of the bed. Make sure resident is in an upright sitting position (at a 90-degree angle).
7. Adjust bed height to where you will be able to sit at resident's eye level. Lock bed wheels.
8. Place the tray where it can be easily seen by the resident, such as on the overbed table.
9. Help the resident to clean her hands if she cannot do it herself.
10. Help resident to put on clothing protector if desired.

Feeding a resident

11. Sit in a chair facing the resident at the resident's eye level. Sit on the stronger side if the resident has one-sided weakness.
12. Tell the resident what foods and beverage are on the tray. Offer a drink of beverage. Ask resident what she would like to eat first.



Feeding a resident

13. Check the temperature of the food. Using utensils, offer the food in bite-sized pieces. Tell the resident the content of each bite of food offered. Alternate types of food, allowing for resident's preferences. Do not feed all of one type before offering another type. Make sure resident's mouth is empty before next bite or sip. Report any swallowing problems to the nurse immediately.



Feeding a resident

14. Offer sips of beverage to the resident throughout the meal. If you are holding the cup, touch it to the resident's lips before you tip it. Give small, frequent sips.
15. Talk with the resident during the meal. Do not rush the resident.
16. Wipe food from resident's mouth and hands as needed during the meal. Wipe again at the end of the meal.



Feeding a resident

17. Remove clothing protector if used. Place it and used washcloths or wipes in the proper containers
18. Remove the food tray. Check for eyeglasses, dentures, or any personal items before removing tray. Place tray in proper area.
19. Make resident comfortable. Keep resident in the upright position for at least 30 minutes. Make sure the bed is free from crumbs.
20. Return bed to lowest position. Remove privacy measures.
21. Place call light within resident's reach.
22. Wash your hands.
23. Report any changes in resident to the nurse.
24. Document procedure using facility guidelines.

7. Demonstrate how to assist with eating

REMEMBER:

NAs may be asked to track the amount of food a resident is eating. Methods of measuring food intake differ from facility to facility. NAs should know which method to use and how to use it.

8. Identify signs and symptoms of swallowing problems

Define the following term:

dysphagia

difficulty swallowing.

8. Identify signs and symptoms of swallowing problems

Any of the following can cause dysphagia:

- Stroke
- Head/neck cancer
- Multiple sclerosis
- Parkinson's disease
- Alzheimer's disease

8. Identify signs and symptoms of swallowing problems

The following are signs and symptoms of swallowing problems and must be reported to the nurse:

- Coughing during or after meals
- Choking during meals
- Dribbling saliva, food, or fluid from the mouth
- Having food residue inside the mouth or cheeks during and after meals
- Gurgling during or after meals or losing voice
- Eating slowly
- Avoiding eating

8. Identify signs and symptoms of swallowing problems

Signs and symptoms of swallowing problems (cont'd):

- Spitting out pieces of food
- Swallowing several times per mouthful
- Clearing the throat frequently during and after meals
- Watering eyes when eating or drinking
- Food or fluid coming up into the nose
- Making a visible effort to swallow
- Breathing rapidly while eating or drinking
- Difficulty chewing food
- Difficulty swallowing medications

8. Identify signs and symptoms of swallowing problems

NAs should know these facts about thickened liquids:

- Thickening improves the ability to control fluid in the mouth and throat.
- A doctor orders the necessary thickness after evaluation by a speech-language pathologist.
- Some beverages arrive already thickened.
- NAs cannot offer residents who must have thickened liquids regular liquids, including water.

8. Identify signs and symptoms of swallowing problems

The three thickened consistencies are

- Nectar thick - the thickness of pear nectar or tomato juice
- Honey thick - the thickness of honey; usually consumed with a spoon
- Pudding thick - the thickness of pudding; must be consumed with a spoon

Transparency 8-3: Preventing Aspiration

- Position in a straight, upright position (90-degree angle) for eating or drinking.
- Offer small pieces of food or small spoonfuls of pureed food.
- Feed the resident slowly.
- Place food in the unaffected side of the mouth.
- Make sure mouth is empty before offering the next bite of food or sip of drink.
- Have residents remain upright for at least 30 minutes after eating and drinking.

8. Identify signs and symptoms of swallowing problems

Define the following terms:

parenteral nutrition (PN)

the intravenous infusion of nutrients administered directly into the bloodstream, bypassing the digestive system.

nasogastric tube

a feeding tube that is inserted into the nose and goes to the stomach.

percutaneous endoscopic gastrostomy (PEG) tube

a feeding tube placed in the stomach through the abdominal wall.

gastrostomy

a surgically-created opening into the stomach in order to insert a tube.

8. Identify signs and symptoms of swallowing problems

REMEMBER:

An NA's role in tube feedings is observation for problems and changes in the resident. NAs never insert or remove tubes, do the feeding, or clean the tubes.

8. Identify signs and symptoms of swallowing problems

Guidelines for tube feedings are as follows:

- Wash hands before assisting with tube feedings.
- Make sure tubing is not coiled or kinked.
- Be aware of NPO orders.
- Report if the tube comes out.
- Doctor prescribes feedings. They will be in liquid form and served at room temperature.
- Head of the bed should always remain elevated at 30 degrees, but during feedings the head of the bed should be elevated at least 45 degrees. Resident should stay upright after eating as long as ordered (at least 30 minutes).
- Give careful skin care.

8. Identify signs and symptoms of swallowing problems

NAs should observe for and report the following regarding tube feedings:

- Redness/drainage around opening
- Skin sores or bruises
- Cyanotic skin
- Resident complaints of pain or nausea
- Choking or coughing
- Vomiting

8. Identify signs and symptoms of swallowing problems

Observe and report, tube feedings (cont'd):

- Diarrhea
- Swollen abdomen
- Fever
- Tube falls out
- Problems with equipment
- Sound of feeding pump alarm
- Change of resident's inclined position

9. Describe how to assist residents with special needs

REMEMBER:

Residents with certain conditions or diseases, such as stroke, Parkinson's disease, Alzheimer's disease or other dementias, head trauma, blindness or confusion may need special assistance when eating.

9. Describe how to assist residents with special needs

NAs should remember these helpful dining techniques for assisting residents with special needs:

- Use assistive devices as ordered.
- Use physical cues like placing your hand over his.
- Use short, clear verbal cues (e.g., “Pick up your spoon” and “Put some carrots on your spoon”). Wait until one task is finished before giving the next cue.
- For visually impaired residents, use imaginary clock face to explain position of food on plate.

9. Describe how to assist residents with special needs

Helpful dining techniques for assisting residents with special needs (cont'd):

- For residents who have had CVA, place food in unaffected or stronger side of mouth. Make sure food is swallowed.
- If resident has blind spots, place food in field of vision.
- If resident has tremors, use physical cues and place food and drinks close to resident.
- Place residents with poor balance in dining room chair with armrests. If resident leans, ask her to keep elbows on the table.

9. Describe how to assist residents with special needs

Helpful dining techniques for assisting residents with special needs (cont'd):

- If resident has poor neck control, neck brace may be used to stabilize head. If resident is in a geri-chair, a wedge cushion behind the head and shoulders may be used.
- If resident bites utensils, ask him to open his mouth. Wait until jaw relaxes to remove utensil.
- If resident pockets food in cheeks, remind him to chew and swallow. Touch cheek. Ask him to use his tongue to get the food.
- If resident holds food in mouth, ask her to chew and swallow. Gently press down on tongue when removing spoon from the mouth to help trigger swallowing. Make sure resident has swallowed before offering more food.



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